Maths Exploring the Terminal Assessment - 05

QCM A · order and repeat

After S25; QCM B after S35.

For each question, only one proposal is accurate.
consult the correction.

1. A three-digit code selected from 0, 1, 2, 3, with authorized repetition, offers:

A. 4×3 B. 3^4

C. 4^3 D. $4!$

2. Two distinct functions are assigned to two out of five persons.
is:

A. 10 B. 20

C. 25 D. 120

3. The number of permutations of four distinct objects is:

A. 16 B. 12

C. 4 D. 24

4. For two disjointed finite sets $A, B, |A \cup B|$ is valid for:

A. $|A| + |B|$ B. $|A||B|$

C. $|A| - |B|$ D. $|A \cap B|$

5. With two cards A and B, the lists of length 2 with repetition are:

A. AB, BA B. AA, BB

C. AA, AB, BA, BB D. A, B

QCM B · complementary and double counting

For each question, only one proposal is accurate.
consult the correction.

6. Choosing three out of eight students with no assigned function includes:

A. 8^3 B. $\binom{8}{3}$

C. $8 \times 7 \times 6$ D. 3^8

7. The number of parts of a five-element set is:

A. $5!$ B. 10

C. 25 D. 32

8. Pascal's relationship gives $\binom{6}{3}$ equal to:

A. $\binom{5}{2}+\binom{5}{3}$ B. $\binom{5}{2}\binom{5}{3}$

C. $2\binom{6}{2}$ D. $\binom{6}{4}+1$

9. One urn contains 3 distinct reds and 2 distinct blues.
red number of:

A. 6 B. 10

C. 9 D. 3

10. Six people are divided into two groups of three. The number of distributions is:

A. 10 B. 20

C. 40 D. 6!

True or false · expected justification. For each statement, indicate whether it is true or false, and then write evidence or a counter-example.

1. $\binom{n}{0}=0$.

2. Counting by supplementary may avoid distinguishing several cases.

3. Shooting three objects simultaneously always gives an order to the objects.